

SQL Query Structure

Components of SQL Queries
(BigQuery)



SQL Query Structure (Part 1)

Clause	Description	Example
SELECT	Columns to retrieve from a table	SELECT Column 1, Column 2 FROM Table;
FROM	Specifies the table to query data from	SELECT * FROM `dataset.table` ;
JOIN	Combines rows from two or more tables based on a related column	SELECT * FROM orders JOIN customers ON orders.cust_id = customers.id;



SQL Query Structure (Part 2)

Clause	Description	Example
WHERE	Filters rows based on a condition(s)	SELECT * FROM employees WHERE salary > 5000;
GROUP BY	Groups rows sharing a value so aggregate functions can be applied	SELECT city, COUNT(*) FROM employees GROUP BY city;
HAVING	Filters aggregations (like WHERE but for aggregations)	SELECT city, COUNT(*) FROM employees GROUP BY city HAVING COUNT(*) > 5;
ORDER BY	Sorts results in ascending (ASC) or descending (DESC) order	SELECT * FROM employees ORDER BY salary DESC;
LIMIT	Restricts the number of rows returned in the result	SELECT * FROM employees LIMIT 10;

SQL Operators

The most common operator
categories



Comparison Operators

Operator	Description	Example
=	Equal to	WHERE city = 'Manama'
!= or <>	Not equal to	WHERE city <> 'Dubai'
>	Greater than	WHERE salary > 5000
<	Less than	WHERE age < 30
>=	Greater than or equal	WHERE grade >= 80
<=	Less than or equal	WHERE marks <= 60



String Operators

Operator	Description	Example
LIKE	Pattern matching (Case Sensitive)	WHERE name LIKE 'Ali'
NOT LIKE	Excludes pattern	WHERE name NOT LIKE 'Ali'



Wildcards Used in LIKE

Operator	Description	Example
%	Represents zero or more characters	'Manama%' → matches “Manama”, “Manamacity”, “Manama123”, etc.
_	Represents exactly one character	'_anama' → matches “Manama”, “Canama”, “Danama”



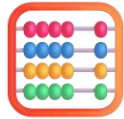
Special Operators

Operator	Description	Example
IN	Match any value in a list	WHERE city IN ('Manama','Riyadh','Doha')
BETWEEN	Range check (inclusive)	WHERE salary BETWEEN 3000 AND 7000
IS NULL	Checks for null values	WHERE email IS NULL
IS NOT NULL	Checks for non-null values	WHERE phone IS NOT NULL



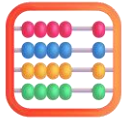
Logical Operators

Operator	Description	Example
AND	All conditions must be true	WHERE city='Manama' AND age>25
OR	Any condition can be true	WHERE city='Manama' OR city='Dubai'
NOT	Negates a condition	WHERE NOT city='Manama'



Arithmetic Operators

Operator	Description	Example
+	Addition	SELECT price + tax FROM products;
-	Subtraction	SELECT price - discount FROM products;
*	Multiplication	SELECT quantity * price FROM sales;
/	Division	SELECT total / count FROM report;
MOD ()	Modulus (remainder) via function	SELECT MOD(10, 3); → 1



Aggregations Functions

Function	Description	Example
COUNT()	Counts the number of rows	COUNT(*)
SUM()	Adds all numeric values	SUM(salary)
AVG()	Calculates the average	AVG(age)
MIN()	Finds the smallest value	MIN(price)
MAX()	Finds the largest value	MAX(price)



Select Functions

Feature	Purpose	Example
ROUND()	Rounds numeric values	ROUND(SUM(sales) , 2)
DISTINCT	Removes duplicates from results	SELECT DISTINCT city



Date Functions

Operator	Description	Example
EXTRACT()	Extract a numeric part: Extract year = 2025	SELECT EXTRACT(YEAR FROM order_date) AS year
DATE_TRUNC() C()	Round/truncate to a boundary: Get start of month 2025-10-01 00:00:00	SELECT DATE_TRUNC(order_date, MONTH) AS month



Date Functions

Common parts for `EXTRACT ()`

Field	Description
YEAR	Extracts year
MONTH	Extracts month number (1–12)
DAY	Extracts day of month
DAYOFWEEK	Day of week (1 = Sunday, 7 = Saturday)
QUARTER	Extracts quarter (1–4)

Common parts for `DATE_TRUNC ()`

Field	Description
YEAR	Start of the year (Jan 1)
QUARTER	Start of the quarter
MONTH	Start of the month
WEEK	Start of week (Sunday; WEEK(MONDAY) for Mon)
DAY	Removes time (00:00:00)

Questions?

Keep this reference handy while you
query away!